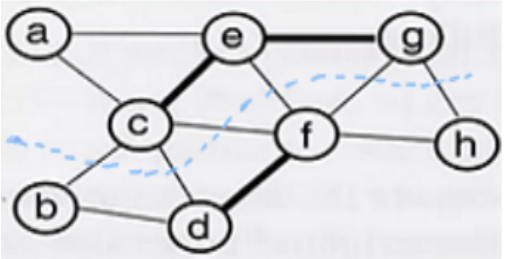
BRAC UNIVERSITY Inspiring Excellence	**BRAC University** **Department of Computer Science and Engineering** Semester: **Spring 2025** Course Code: **CSE460** Course Title: **VLSI Design**	**Final Exam** Full Marks: **15 x 3 = 45** Time: **1 hour 30 minutes** Date: **18th March 2025**

Set A

Student ID:	**Name:**	**Section:**

[Answer any **THREE** questions out of **FOUR**. Each question carries **15 marks**. Return the question paper with the answer script.]

1. (CO4) Figure 1 shows an initial partitioning of 8 nodes (**a, b, c, d, e, f, g, h**).
The cut line divides the circuit elements into two partitions.



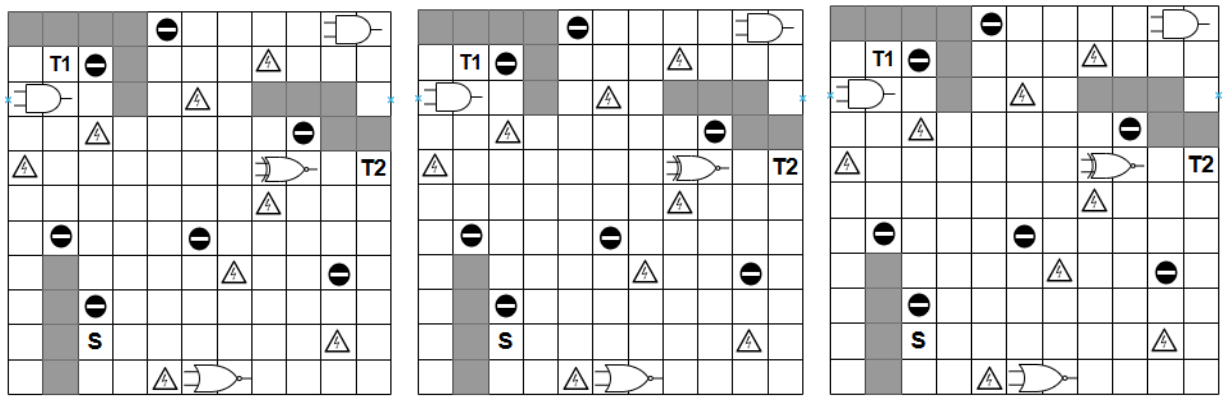
(a) **What** is the initial cut cost? **Compute** the costs of moving each node from one partition to another. **[1+1]**

(b) **Perform** the first iterations of the first pass of the KL algorithm **[6]**
i. Find the maximum gain of swapping a pair of nodes (Δg_1) in the first iteration.
ii. Swap the pair and draw the updated partition after the first iteration.

(c) Now **perform** the second iteration of the first pass of the algorithm on the updated partition you got in 1(b) and find the maximum gain (Δg_2) for the second iteration. **[4]**

(d) If the maximum gain in the next 2 iterations is $\Delta g_3 = -\Delta g_1$ and $\Delta g_4 = -\Delta g_2$, then after how many swaps do we get the optimized partition? What will be the maximum cumulative gain (Gm) and the new cut cost? **[3]**

2. (CO4) For question (a), you are given a certain chip planning task. You have four blocks with the following widths and
heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 9, h = 1 or w = 3, h = 3
For questions (b-d), refer to the figures at the bottom of the question. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown. **Gates are not obstacles**.



Metal	Obstacle 1	Obstacle 2

	(a)	**Find** and **draw** a floor plan with the minimum total area enclosed if the global bounding box is a 10x10 grid.	**[3]**
	(b)	Using the modified Lee's Maze Algorithm, **find** the shortest path from S to T1 and T2, allowing minimum bends. (Use the sequence 123123…….)	**[4]**
	(c)	**Find** the total memory usage of the given grid for the sequence mentioned in (b).	**[4]**
	(d)	If the maximum memory requirement of the provided grid is 32 bytes, **find** how many unique numbers starting from 0 can be used for wave propagation in Lee's maze algorithm and suggest a full sequence.	**[4]**

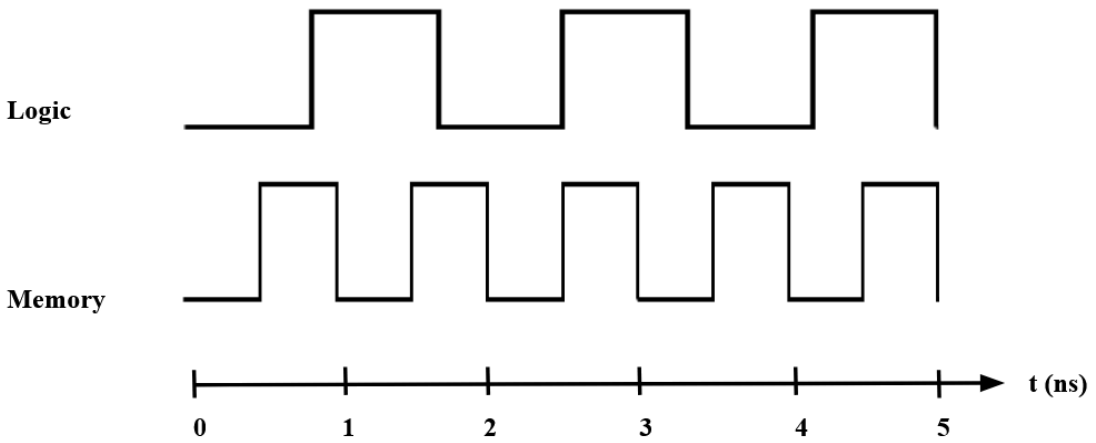
3. (CO3)

Suppose the output of a 4-input NAND gate is connected to two logic gates. One of the logic gates inverts the signal, and the other gate gives a HIGH output only when all three of its inputs are LOW. Consider that we will design for equal fall and rise times. For (b), (c), and (d), assume $\mu_n = 2\mu_p$.

(a)	Assuming $\mu_n = 10\mu_p$., **find** k_n and k_p for a NOR-2 and a NOR-5000 gate.	**[1+2]**
(b)	**Identify** the gates used as loads in the stem and calculate the value of C_{load} in terms of C.	**[1+2]**
(c)	**Draw** the RC equivalent circuit of the 4-input NAND gate. Include C_{load} found in (b).	**[4]**
(d)	Use the Elmore delay model to **find** the expressions for tcdr, tpdr, tcdf, and tpdf	**[5]**

4. (CO3)

Suppose you are designing a chip in a 1.1 V, 45 nm technology which has 50 million transistors, of which 20 million are used for building logic gates and the rest are used in memory arrays. The average logic transistor width is 10λ and the average memory transistor width is 7λ. The chip has an area of 60 mm^2. The gate and diffusion capacitances are 15 pF/mm^2 and 80 pF/mm^2, respectively for a total 50 million transistors. Neglect the wire capacitance. The operating frequency is 5 GHz. The switching signals for both logic and memory transistors are shown in figure 4 (a).



You also obtain the following current data for each transistor:

- Gate leakage is 5 nA/μm.
- Junction leakage is 2 nA/μm
- Subthreshold leakage is 10 nA/μm

The acceptable total power consumption of a chip is 10 W.

(a)	**Find** the activity factor of the logic gates and the memory arrays.	**[2]**
(b)	**Calculate** the switching power consumption of the chip.	**[6]**
(c)	**Estimate** the static power of the chip. Thereafter, **calculate** the total power consumption.	**[3]**
(d)	Is the total power consumption within the acceptable range? If not, find the maximum clock frequency to keep the total power within the acceptable range.	**[4]**